

FINAL — UGV Software-Validation Prototype

Locked Build Document — matches the final corrected wiring schematic exactly

This revision supersedes all earlier drafts. It reflects the corrected diagram in which Front HC-SR04 VCC is 5V, the GPIO33/34 conflict is resolved, all four Echo dividers are explicit, and rail capacitors are explicitly placed.

STATUS: FROZEN FOR BUILD. Architecture, GPIO map, and wiring are final. Remaining items below are purchase-time verifications only, not design changes.

1. Final Architecture



2WD rear-wheel drive with front Ackermann steering. Rear motors propel; MG90S mechanically steers the front wheels. No differential-speed steering is used.

2. Final Bill of Materials (with prices)

2.1 Core electronics and sensors

#	Component / model	SKU	Qty	Unit ■	Total ■
1	ESP32-WROOM-32 DevKit	835800	1	395	395
2	ESP32-CAM, OV2640 — confirm delivered sensor label	456154	1	759	759
3	ESP32-CAM-MB Micro-USB download module	1150750	1	109	109
4	MPU-6050 IMU breakout	2846	1	151	151
5	HC-SR04 ultrasonic sensor	2444	4	69	276
6	MG90S 180° steering servo	1308083	1	125	125
7	1kOhm 1/4W resistor	local/Robu	10	2	20
8	2kOhm 1/4W resistor	local/Robu	10	2	20

2.2 Drive, power, and safety

#	Component / model	SKU	Qty	Unit ■	Total ■
9	TT DC motor pair, 1:48, 3-6V, 200RPM at 6V	1121374	1 pair	223	446
10	TB6612FNG dual motor driver breakout	Robu product	1	489	489
11	XL4015 5A buck with heatsink — verify input range covers 7.4V	1510171/R2246 23	1	71	71

#	Component / model	SKU	Qty	Unit ■	Total ■
12	LM2596S adjustable buck, genuine 3A	11548	1	44	44
13	Samsung INR18650-35E 2S1P 7.4V 3500mAh 2C battery	1366608	1	1075	1075
14	5A automotive blade fuse + inline holder	local/automotive	1 set	80	80
15	NC emergency-stop switch, DC-rated >=5A — verify true NC contact	local/Robu	1	120	120
16	Rocker switch, DC-rated	R132455	1	6	6
17	Power distribution terminal block	local	1	100	100
18	5V polyfuse/fuse (2-3A) for electronics branch	local	1	40	40
19	2S Li-ion charger, 8.4V CC/CV — NEW, previously missing	local/Robu	1	350	350

Item 19 (2S charger) is a new addition. It was absent from all earlier BOM versions and from the wiring diagram, which shows discharge only. Do not use a single-cell 4.2V charger on this 2-cell pack.

2.3 Mechanics, filtering, wiring

#	Component	Qty	Est. ■
20	65-70mm TT/BO compatible wheels	4	150
21	3-5mm acrylic sheets	2	400
22	M3 hardware set	1 set	200
23	Servo horn, tie rods, steering knuckles, axles/bearings	1 set	200
24	470uF 16V electrolytic capacitors	2	30
25	100uF 10-16V electrolytic capacitors	2	20
26	0.1uF 50V ceramic capacitors	8	30
27	Perfboard ~5x7cm	1	50
28	18-20 AWG stranded wire (~3m)	1 lot	120
29	22-24 AWG stranded wire (~3m)	1 lot	80
30	Dupont M-M/M-F/F-F set	1 set	150
31	Heat shrink + zip ties	1 set	120
32	25mm checkerboard print	1	50

Revised estimated total: approximately ■6,166 (previous total plus the newly added 2S charger).

3. Full Power Wiring

3.1 Battery positive chain

Battery + (7.4V nominal / 8.4V full)
-> 18-20 AWG red -> 5A blade fuse holder -> 5A blade fuse
-> E-stop terminal 1 (NC) -> E-stop terminal 2
-> Rocker switch terminal 1 -> Rocker switch terminal 3
-> Power Distribution Terminal Block (+ BUS)
-> Branch 1: XL4015 IN+
-> Branch 2: LM2596 IN+

3.2 Common ground bus

Battery - (black) -> Power Distribution Terminal Block (- BUS) -> COMMON GROUND
-> XL4015 IN-/OUT-
-> LM2596 IN-/OUT-
-> TB6612FNG GND
-> ESP32-WROOM GND
-> ESP32-CAM GND
-> MG90S brown wire
-> HC-SR04 GND x4
-> MPU-6050 GND
-> capacitor negative legs

3.3 6V motor rail

From	To	Setting
+BUS	XL4015 IN+	—
XL4015 OUT+	TB6612 VM	Set to exactly 6.00V with multimeter before connecting
XL4015 OUT-	Common GND	—
470uF capacitor	Across TB6612 VM to GND	Bulk filtering, short leads

3.4 5V electronics rail

From	To	Setting
+BUS	LM2596 IN+	—
LM2596 OUT+	5V Fuse (2-3A) -> 5V distribution	Set to exactly 5.00V with multimeter before connecting
LM2596 OUT-	Common GND	—
5V distribution	ESP32-WROOM 5V, ESP32-CAM 5V, MG90S red, HC-SR04 VCC x4	—
470uF + 100uF	Across 5V distribution to GND	Bulk filtering, short leads

4. Full Signal Wiring

4.1 ESP32-WROOM GPIO map (FINAL, matches diagram exactly)

GPIO	Connects to	Function
4	TB6612 BIN2	Right motor direction 2
5	Front HC-SR04 TRIG	Front trigger
13	TB6612 PWMA	Left motor PWM
14	TB6612 PWMB	Right motor PWM
16	Front HC-SR04 ECHO (via divider)	Front distance input
17	Left HC-SR04 TRIG	Left trigger
18	MG90S signal	Steering PWM
19	Right HC-SR04 TRIG	Right trigger
21	MPU-6050 SDA	I2C data
22	MPU-6050 SCL	I2C clock
23	TB6612 STBY	Firmware motor enable/disable
25	TB6612 BIN1	Right motor direction 1
26	TB6612 AIN1	Left motor direction 1

GPIO	Connects to	Function
27	TB6612 AIN2	Left motor direction 2
32	Left HC-SR04 ECHO (via divider)	Left distance input
33	Rear HC-SR04 TRIG	Rear trigger
34	Right HC-SR04 ECHO (via divider)	Right distance input, input-only pin
35	Rear HC-SR04 ECHO (via divider)	Rear distance input, input-only pin

4.2 TB6612FNG pins

Pin	Connects to
VM	XL4015 OUT+ (6.00V)
VCC	ESP32 3.3V
GND	Common GND
STBY	ESP32 GPIO23
PWMA	ESP32 GPIO13
AIN1	ESP32 GPIO26
AIN2	ESP32 GPIO27
AO1/AO2	Left TT motor
PWMB	ESP32 GPIO14
BIN1	ESP32 GPIO25
BIN2	ESP32 GPIO4
BO1/BO2	Right TT motor

4.3 MG90S / MPU-6050 / HC-SR04 (all four)

Module	Pin	Connects to
MG90S	Brown	Common GND
MG90S	Red	5V distribution
MG90S	Orange	GPIO18
MPU-6050	VCC	ESP32 3.3V
MPU-6050	GND	Common GND
MPU-6050	SDA	GPIO21
MPU-6050	SCL	GPIO22
MPU-6050	AD0	GND
MPU-6050	INT	Not connected
Front HC-SR04	VCC	5V distribution (CORRECTED to 5V)
Front HC-SR04	TRIG/ECHO	GPIO5 / GPIO16 via divider
Left HC-SR04	VCC	5V distribution
Left HC-SR04	TRIG/ECHO	GPIO17 / GPIO32 via divider
Right HC-SR04	VCC	5V distribution
Right HC-SR04	TRIG/ECHO	GPIO19 / GPIO34 via divider
Rear HC-SR04	VCC	5V distribution
Rear HC-SR04	TRIG/ECHO	GPIO33 / GPIO35 via divider

Echo divider (build x4): HC-SR04 ECHO(5V) -> [1k0hm] -> node -> ESP32 GPIO
node -> [2k0hm] -> GND

5. Mechanical Dimensions

Parameter	Value
Overall length	300 mm
Overall width	200 mm
Wheelbase	220 mm
Track width	180 mm
Wheel diameter	65-70 mm
Ground clearance	30-40 mm
Max steering angle	~25-30 deg
Min turning radius	~400-470 mm

6. Before-First-Power-On Verification (frozen, purchase-time only)

#	Verify	Why
1	Delivered TT motor is genuinely 3-6V; get stall current	Confirms XL4015 sizing and TB6612 load margin
2	XL4015 input range covers 7.4V (not "8-36V only" variant)	2S battery drops below 8V during use; wrong module may stop regulating
3	E-stop has true NC contact block, DC-rated >=5A	Confirms independent hardware safety layer works as intended
4	2S Li-ion charger (8.4V CC/CV) matches battery connector	Battery has no charging path defined until this is purchased
5	XL4015 set to exactly 6.00V with multimeter before connecting TB6612	Prevents overvoltage to motors
6	LM2596 set to exactly 5.00V with multimeter before connecting ESP32/servo/sensors	Prevents overvoltage to logic boards
7	Common ground continuity checked across every module	GPIO signals are unreliable without a shared ground reference

7. Firmware Safety Sequence

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BOOT: STBY=LOW, PWM=0
-> init GPIO directions, MPU-6050, HC-SR04 sensors
-> establish Wi-Fi
-> receive fresh valid command + safety healthy
-> STBY=HIGH, allow PWM

ON FAULT (timeout / invalid command / unsafe sensor / internal fault):
-> PWM=0 -> STBY=LOW -> require fresh valid command to resume

PHYSICAL E-STOP: cuts all power upstream of both converters,
independent of firmware; reboot/reconnect required after release.
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8. Final Sign-Off

This document and its accompanying wiring diagram are now FROZEN for build. All GPIO assignments, power rails, Echo dividers, capacitor placements, and safety logic are final. Only the seven purchase-time verification items above remain before first power-on.